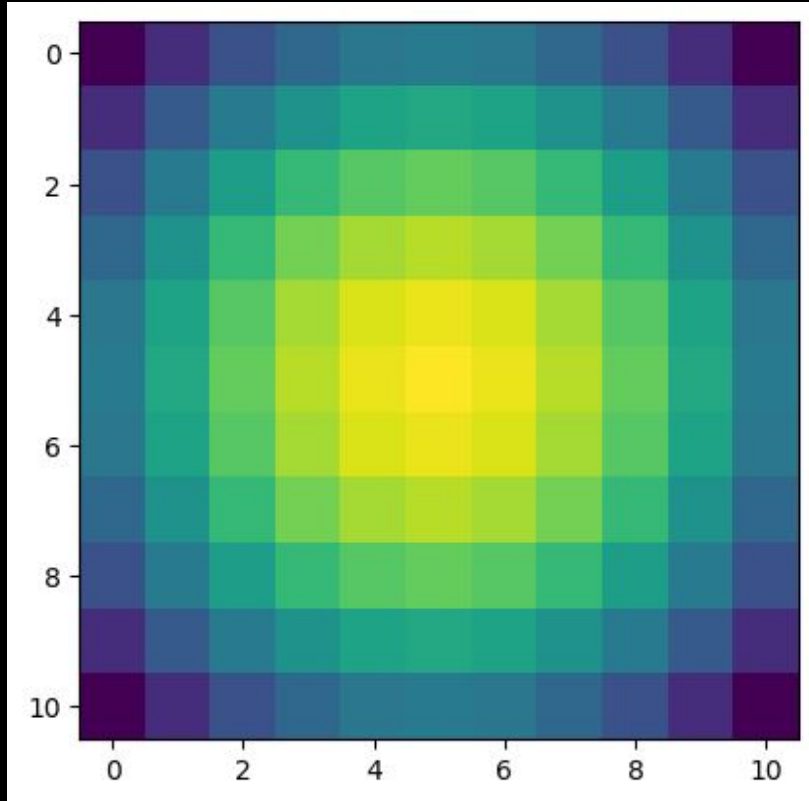


Assignment - Brain MRI Segmentation

Chosen value of $q \rightarrow 2$

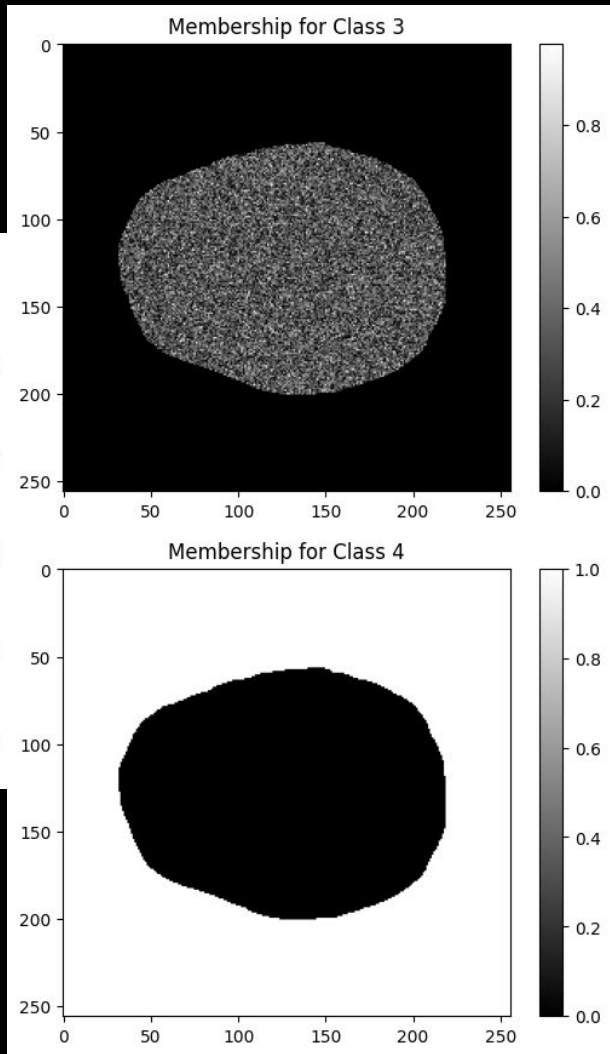
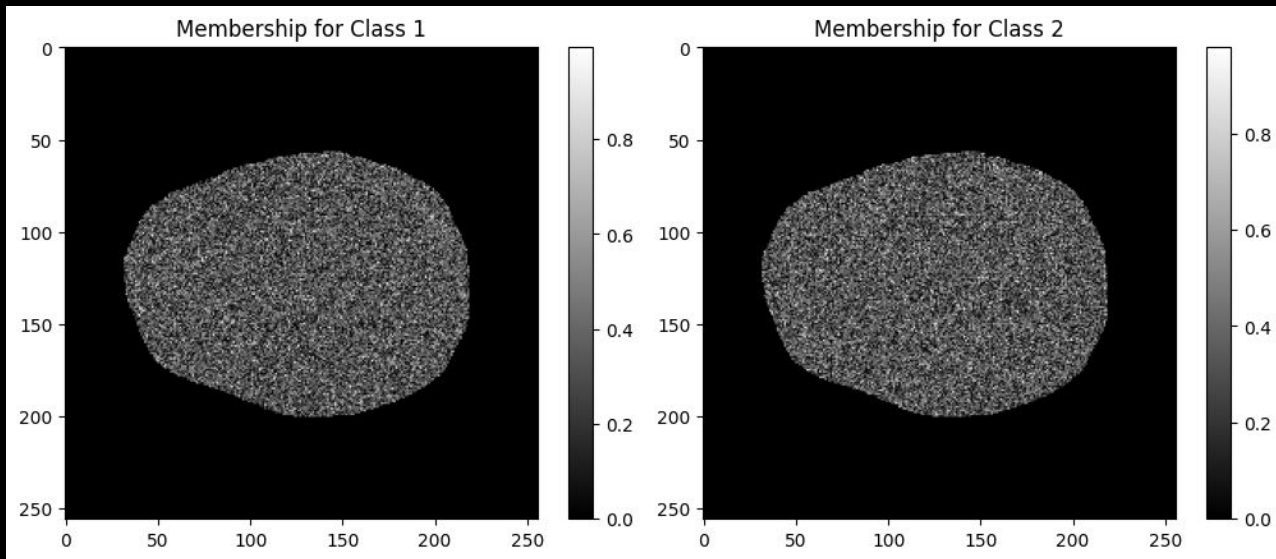
Can try to choose a value lesser than 2 to pick sparser solutions.

Neighborhood mask image



Kept a variable `kernel_size` to control size of neighborhood.
Also can pick variance of kernel according to how smooth the bias is.

Initial estimate of memberships



Initialized memberships for classes 1-3 randomly, class 4 represents the background, hence it is 1 only where mask is 0.

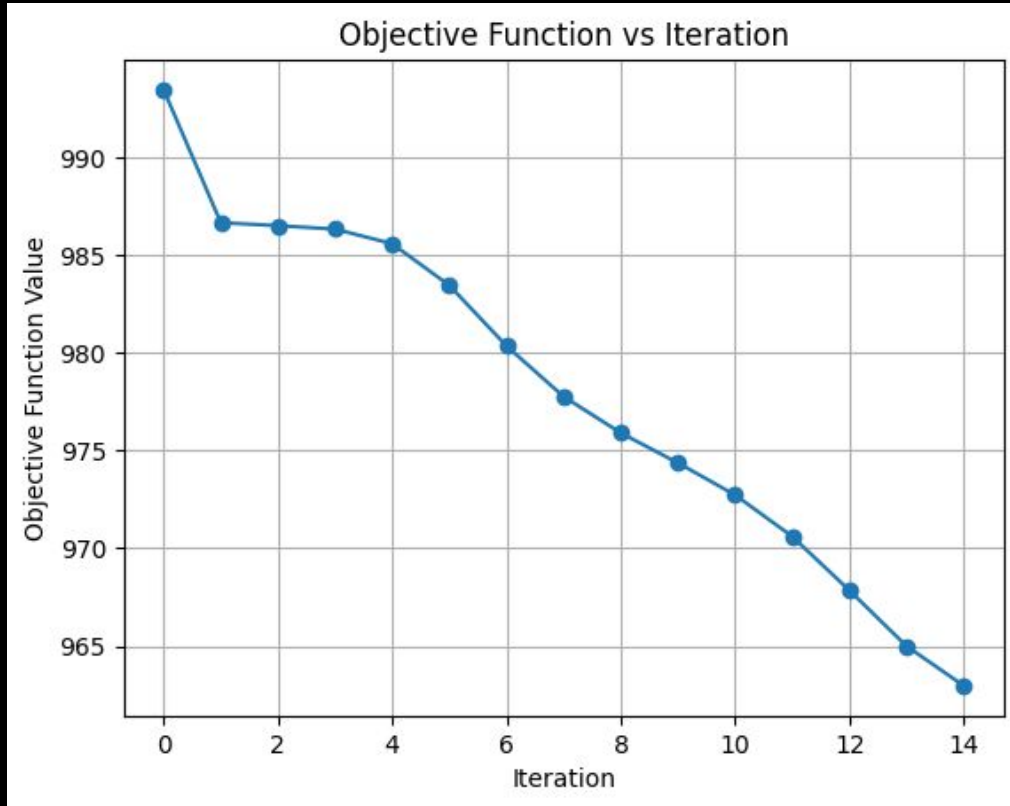
Initial estimate of class means

Class 1	Class 2	Class 3	Class 4
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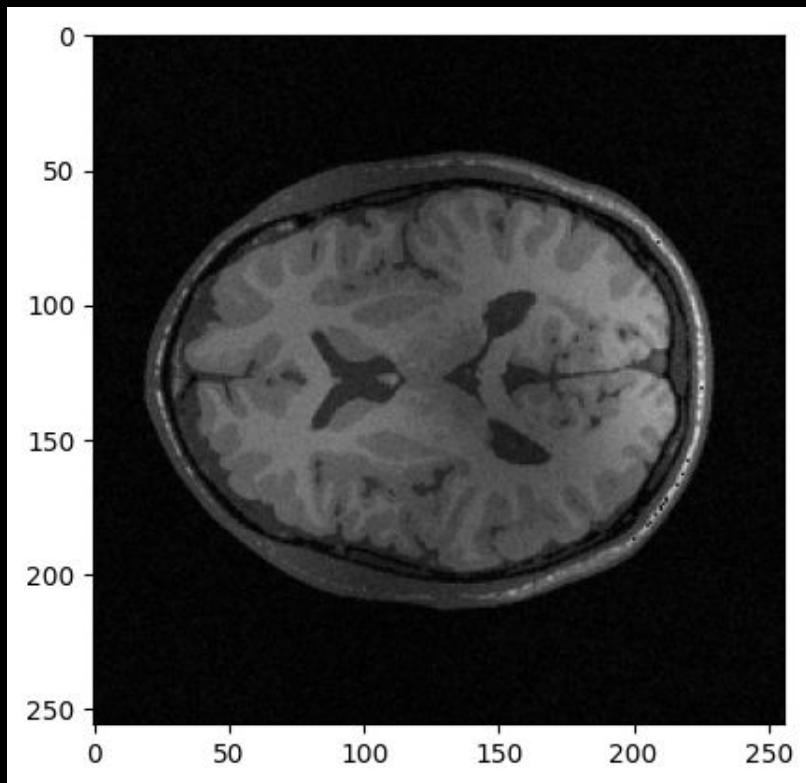
0.06818729	0.50286959	0.65542129	0.0
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Initialized randomly, 4th class represents the background, hence class mean is 0.

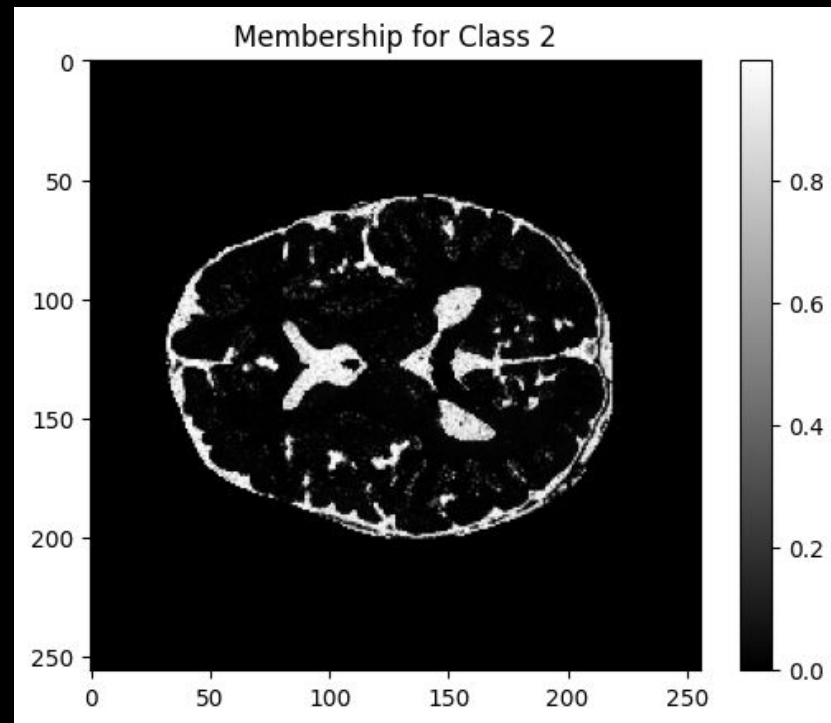
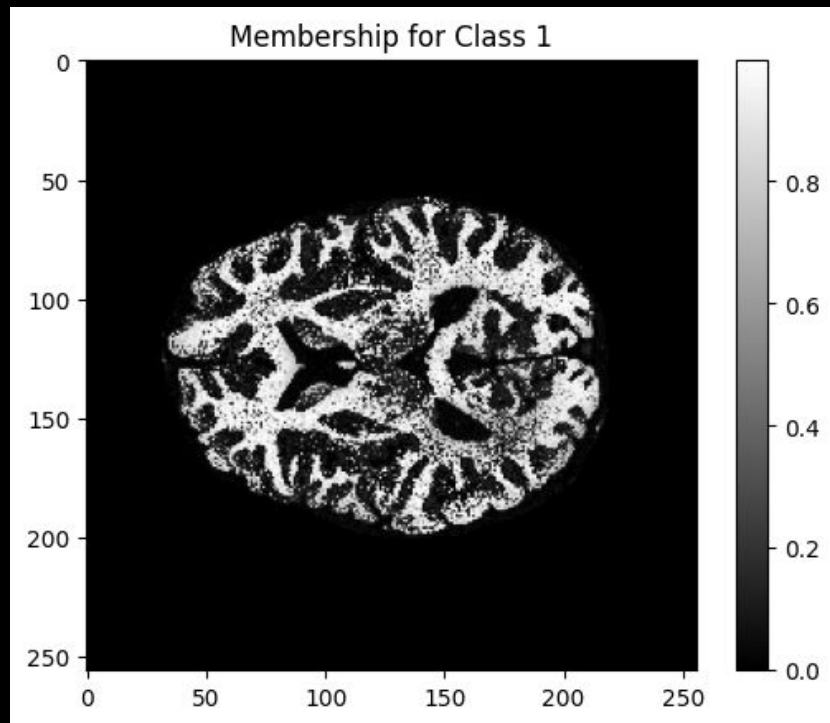
Objective Function vs Iteration



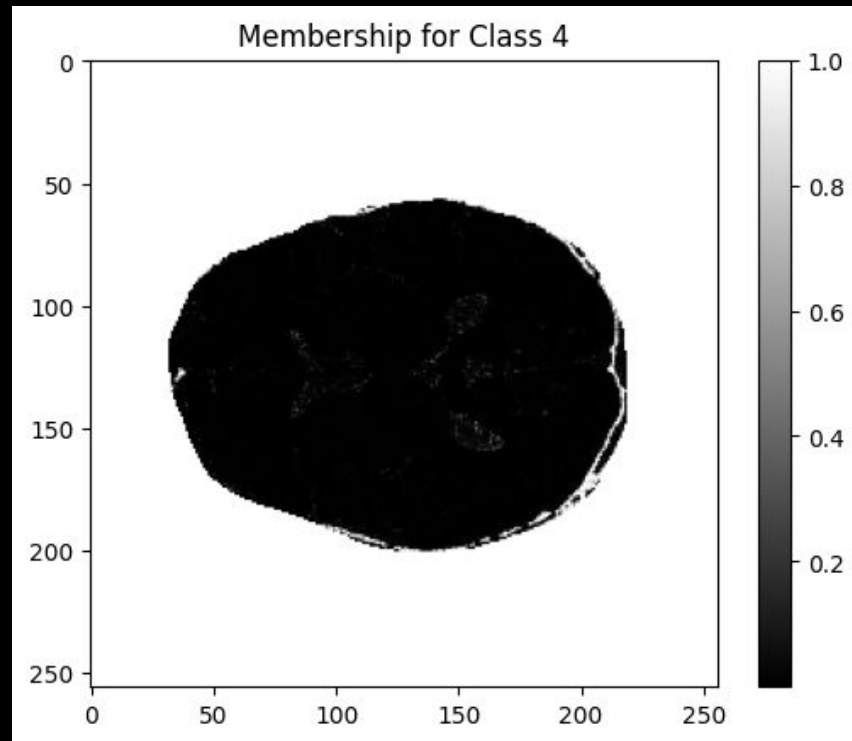
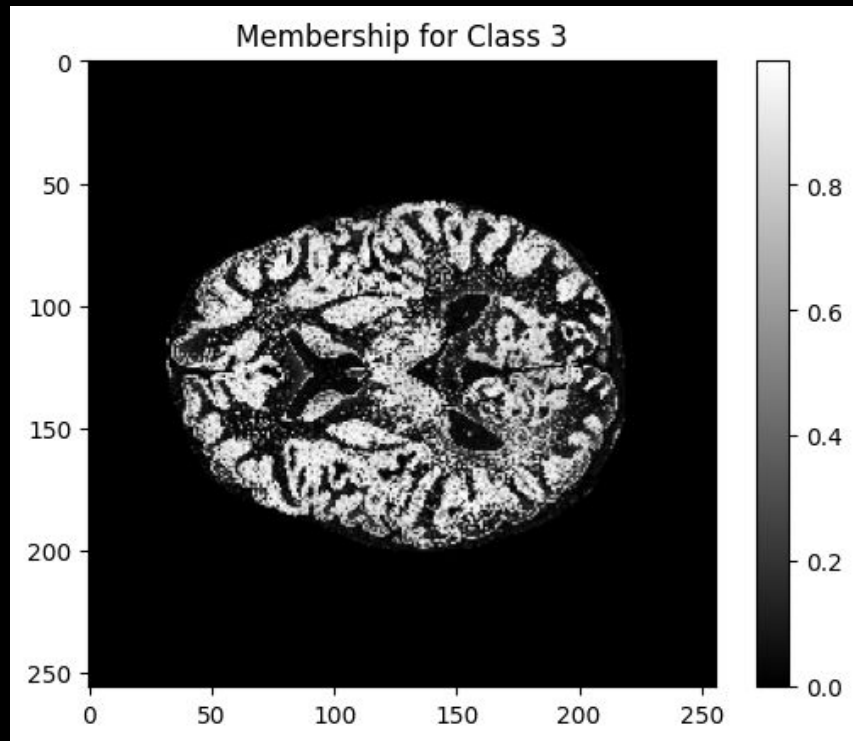
Corrupted Image



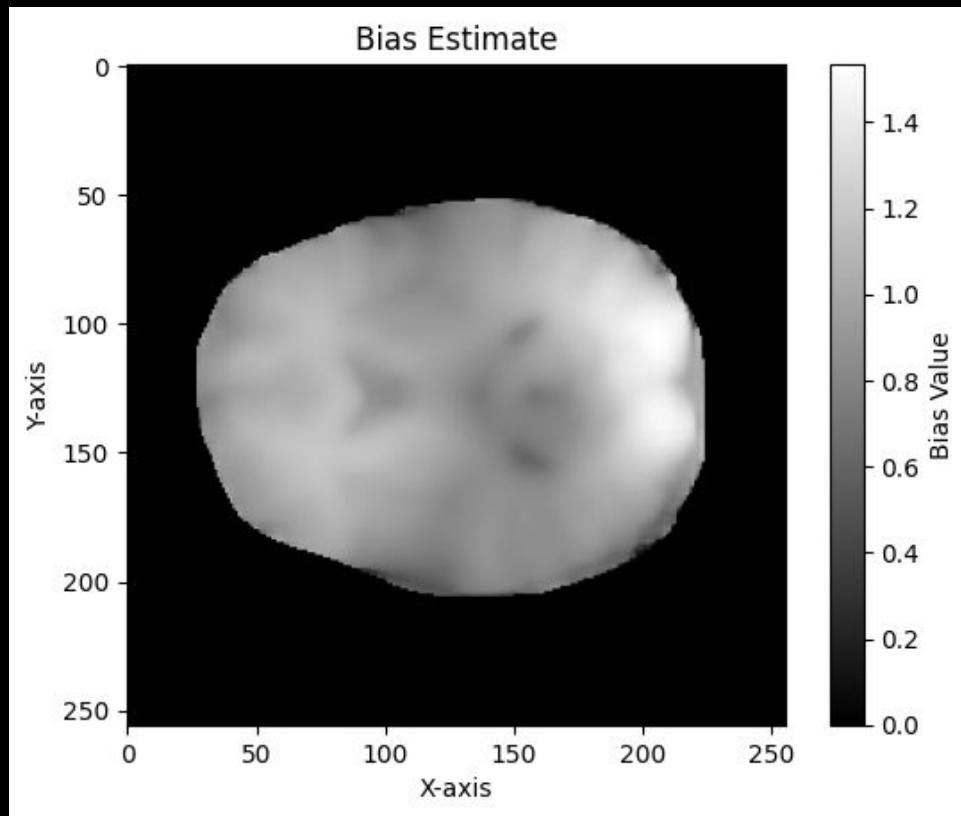
Optimal class membership images



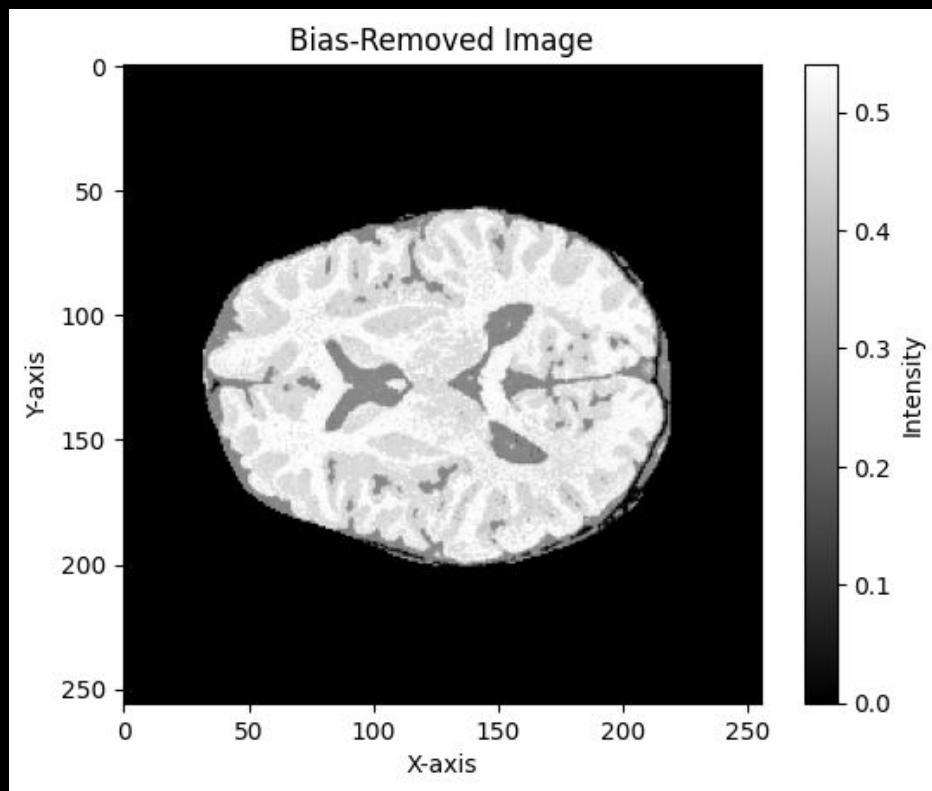
Optimal class membership images



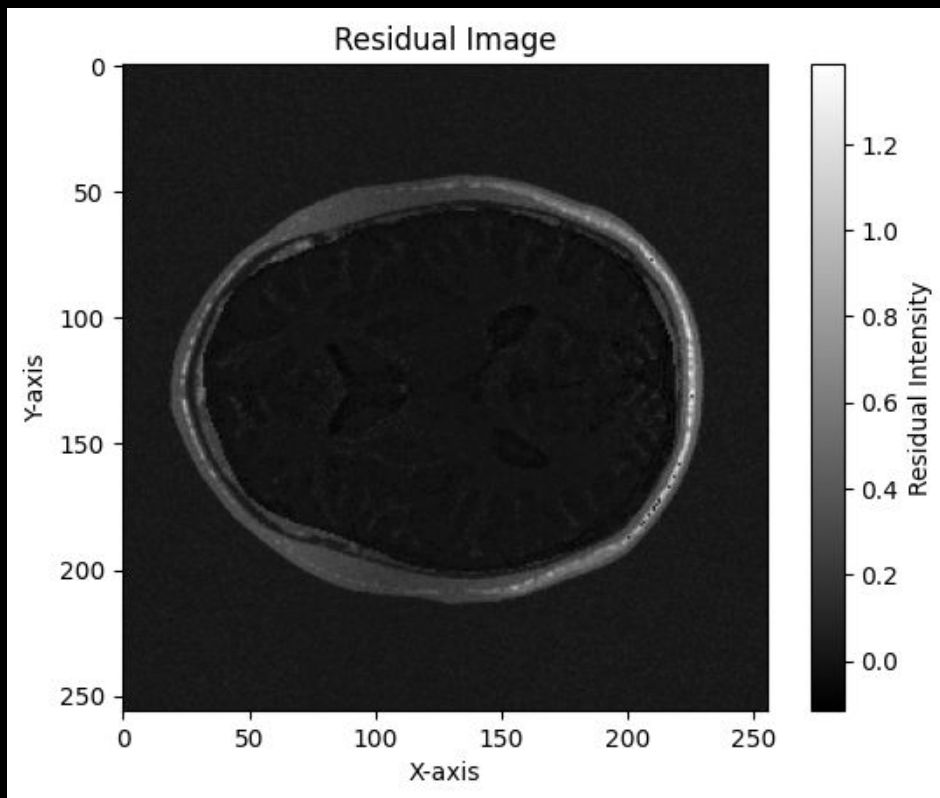
Optimal bias field estimate



Bias-removed image



Residual Image



Optimal class mean estimates

Class 1

Class 2

Class 3

Class 4

0.54030573

0.28553791

0.46064138

0.0

Unique Solution

No, the current approach does not give a unique solution. We can scale the bias up by a factor, scale the class means down by a factor, and that would still remain a solution. This leads to a huge valley of solutions.

To fix this, we can fix the class mean of a class to a constant. That will force the other biases and class means to adjust to it, and give a unique solution.